

Advanced Smart Grocery Store System (2D Array Based)

◆ **Problem Statement (Advanced Version)**

Design a **Smart Grocery Store Management System** that handles multiple categories of products using **2D arrays**, along with **constructors, inheritance, loops, and conditional logic**.

◆ **System Requirements**

◆ **1. Class Structure (OOP + Inheritance)**

- Create a base class Product:
 - Attributes: name, price, quantity
 - Constructor to initialize values
 - Method: calculateTotal()
 - Create derived classes:
 - PerishableProduct
 - Extra attribute: expiryDays
 - Method: checkExpiry()
 - NonPerishableProduct
 - Extra attribute: warrantyMonths
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◆ **2. 2D Array Usage**

Use a **2D array to store products by category**:

Rows → Categories (Fruits, Dairy, Snacks)

Columns → Products in each category

Example:

```
products[3][3]
```

Row 0 → Fruits

Row 1 → Dairy

Row 2 → Snacks

Each cell stores a **Product object**.

◆ 3. Functional Requirements

✓ A. Add Products

- Use constructors to initialize products
 - Store them in the 2D array
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✓ B. Display All Products

- Use **nested loops** to traverse the 2D array
 - Show:
 - Name
 - Price
 - Quantity
 - Category
-

✓ C. Calculate Total Bill

- Loop through all products
 - Use method `calculateTotal()`
 - Sum all values
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✓ D. Apply Smart Discounts (if-else)

- If total bill > 5000 → 10% discount
- If total bill > 10000 → 20% discount
- Otherwise → no discount

✓ E. Expiry Check (Perishable Only)

- If expiryDays < 3
 - Show warning: "Expiring Soon ⚠"
- If expiryDays <= 0
 - Mark as "Expired ❌"

✓ F. Low Stock Alert

- If quantity < 5
 - Show: "Restock Needed 🔔"

✓ G. Category-wise Total (Using 2D Logic)

- Calculate total price **for each row (category)** separately

◆ Advanced Logic Features

- Use **nested loops** for 2D traversal
- Use **instanceof** (Java) to check:
 - Perishable vs Non-perishable
- Use **method overriding**:
 - calculateTotal() can behave differently if needed

◆ Sample Output (Concept)

Category: Fruits

Apple - Price: 200 Qty: 3 → Total: 600

⚠ Expiring Soon

Category: Dairy

Milk - Price: 150 Qty: 2 → Total: 300

✖ Expired

Category: Snacks

Chips - Price: 100 Qty: 10 → Total: 1000

Total Bill: 1900

Discount Applied: 0%

Final Bill: 1900

Low Stock Alert:

Milk needs restocking!